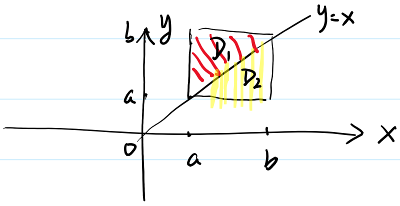
第14讲. 二重积分性质、计算.

作业: P_{103}. 1, 3, 4, 5, 6, 7, 9.

例1. 证明: 若 $f(x) \in C[a,b]$. 且 $f(x)>0$, $\forall x \in [a,b]$.

则 $\iint\limits_D \frac{f(x)}{f(y)} d\sigma \geqslant (b-a)^2$. $D=[a,b]\times[a,b]$.

证:



$$\iint\limits_D \frac{f(x)}{f(y)} d\sigma = \iint\limits_D \frac{f(y)}{f(x)} d\sigma = \frac{1}{2}\iint\limits_D \left(\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\right) d\sigma$$

$$\geqslant \iint\limits_D d\sigma = (b-a)^2.$$

性质1. 若 $f(x,y)$, $g(x,y) \in R(D)$. 则 $f(x,y)g(x,y) \in R(D)$. 且 $\forall T=\{\Delta D_1,\cdots,\Delta D_n\}$.

$\forall (x_i,y_i)\in\Delta D_i$, $\forall (\xi_i,\eta_i)\in\Delta D_i$. 有

$$\lim_{\|T\|\to 0}\sum_{i=1}^{n} f(x_i,y_i)g(\xi_i,\eta_i)\Delta\sigma_i = \iint\limits_D f(x,y)g(x,y)d\sigma$$

定理2. 设 $f(x,y)\in R(D)$. 则 $f(x,y)$ 在 D 上有界. 即 $\exists M>0$. s.t. $\forall (x,y)\in D$.

$$|f(x,y)| \leq M.$$

$$f(x,y)=\begin{cases} 1 & (x,y)\in \underbrace{[0,1]\times[0,1]}_{D} \text{ 有理点对.} \\ 0 & \text{其它} \end{cases}$$

$\forall T=\{\Delta D_1,\cdots,\Delta D_n\}$.

$\forall (x_i,y_i)\in\Delta D_i$. $\sum_{i=1}^{n} f(x_i,y_i)\Delta\sigma_i = 1$

$f(x,y)$ 在 D 上有界. 但 $f(x,y)\notin R(D)$.

设 $f(x,y)$ 在 D 上有界.

$\forall T=\{\Delta D_1,\cdots,\Delta D_n\}$.

$$M_i = \sup\{f(x,y) \mid \forall (x,y)\in\Delta D_i\},$$

$$m_i = \inf\{f(x,y) \mid \forall (x,y)\in\Delta D_i\}$$

$M_i = \sup\{ f(x,y) \mid \forall (x,y) \in \Delta D_i \}$.

$m_i = \inf\{ f(x,y) \mid \forall (x,y) \in \Delta D_i \}$.

$S(T) = \sum_{i=1}^{n} M_i \Delta \sigma_i$ —— $f(x,y)$ 对应于 T 达布上和.

$s(T) = \sum_{i=1}^{n} m_i \Delta \sigma_i$ ～～ ………下和.

定理2. 设 $f(x,y)$ 定义在有界闭区域 D 上. 则 $f(x,y) \in R(D) \Leftrightarrow \forall T = \{\Delta D_1, \cdots, \Delta D_n\}$.

$$\lim_{\|T\| \to 0} S(T) = \lim_{\|T\| \to 0} s(T) \Leftrightarrow \lim_{\|T\| \to 0} \sum_{i=1}^{n} (M_i - m_i) \Delta \sigma_i = 0.$$

$$w_i = M_i - m_i = \sup\{ f(x,y) - f(u,v) \mid (x,y), (u,v) \in \Delta D_i \}.$$

定理3. 若 $f(x,y) \in C(D)$. D 有界闭区域. 则 $f(x,y) \in R(D)$.

性质1 证明: $\forall T = \{\Delta D_1, \cdots, \Delta D_n\}$. $\forall (x_i, y_i), (\xi_i, \eta_i) \in \Delta D_i$.

$\because f(x,y) \in R(D)$. $\therefore \exists M > 0$, s.t. $\forall (x,y) \in D$. $|f(x,y)| \le M$.

$\because g(x,y) \in R(D)$. $\therefore \lim_{\|T\| \to 0} \sum_{i=1}^{n} w_i(g) \Delta \sigma_i = 0$.

$$\therefore 0 \le \left| \sum_{i=1}^{n} f(x_i, y_i) g(\xi_i, \eta_i) \Delta \sigma_i - \sum_{i=1}^{n} f(x_i, y_i) g(x_i, y_i) \Delta \sigma_i \right| \to \iint_D fg \, d\sigma.$$

$$= \left| \sum_{i=1}^{n} f(x_i, y_i) \left(g(\xi_i, \eta_i) - g(x_i, y_i) \right) \Delta \sigma_i \right|$$

$$\le M \sum_{i=1}^{n} w_i(g) \Delta \sigma_i \to 0. \quad (\|T\| \to 0)$$

定理4. (积分中值). 设 $D \subset \mathbb{R}^2$ 有界闭区域. $f(x,y) \in C(D)$. $g(x,y) \in R(D)$. 且 $g(x,y)$ 在 D 上不变号.

则 $\exists (\xi, \eta) \in D$ s.t. $\iint_D f(x,y) g(x,y) \, d\sigma = f(\xi, \eta) \iint_D g(x,y) \, d\sigma$.

设 $g(x,y) \ge 0$. $\forall (x,y) \in D$. $\iint_D g(x,y) \, d\sigma \ge 0$.

证: $\because f(x,y) \in C(D)$

$\therefore M = \max\{ f(x,y) \mid (x,y) \in D \}$. $m = \min\{ f(x,y) \mid (x,y) \in D \}$.

$\forall (x,y) \in D$. $\iint_D m g(x,y) \, d\sigma \le \iint_D f(x,y) g(x,y) \, d\sigma \le \iint_D M g(x,y) \, d\sigma$

1°. $\iint_D g(x,y) \, d\sigma = 0$. ✓

若 $\iint\limits_D g(x,y)\,d\sigma \neq 0$.

$$m \le \frac{\iint\limits_D fg\,d\sigma}{\iint\limits_D g\,d\sigma} \le M$$

$$\frac{\iint\limits_D fg\,d\sigma}{\iint\limits_D g\,d\sigma} = f(\xi,\eta)$$

$\exists(\xi,\eta)\in D$.

$f(x_0,y_0)>0$. $\quad f(x,y)\in C(D)$. $\quad \lim\limits_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0) > \frac{1}{2}f(x_0,y_0)$

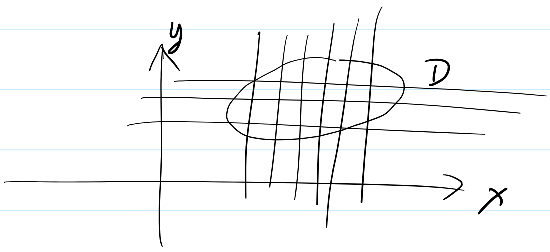
$\exists\delta>0$. s.t. $\forall(x,y)\in B((x_0,y_0),\delta)$. $\quad f(x,y) > \frac{1}{2}f(x_0,y_0)$

二. 计算.

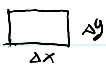
1. 直角坐标系

$$\iint\limits_D f(x,y)\,d\sigma = \lim_{\|T\|\to 0}\sum_{i=1}^{n} f(\xi_i,\eta_i)\Delta\sigma_i$$

$\forall T$.



$\Delta x>0,\ \Delta y>0$.



$$\begin{array}{ccc} \Delta\sigma = & \Delta x & \cdot\ \Delta y \\ \downarrow & \downarrow & \downarrow \\ d\sigma & dx & dy \end{array}$$

$\|T\|\to 0$.

$$\iint\limits_D f(x,y)\,d\sigma = \iint\limits_D f(x,y)\,dx\,dy.$$

定理5. 设 $D=[a,b]\times[c,d]$. $\quad f(x,y)\in R(D)$.

设 $\forall y\in[c,d]$. $\quad I(y)=\int_a^b f(x,y)\,dx$ 存在.

则先对 x 后对 y 累次积分 $\int_c^d\left(\int_a^b f(x,y)\,dx\right)dy$ 存在. 且 $\iint\limits_D f(x,y)\,dx\,dy = \int_c^d\left(\int_a^b f(x,y)\,dx\right)dy$.

证: $\forall[c,d]$. $\quad c=y_0<y_1<\cdots<y_m=d$.

$\forall[a,b]$. $\quad a=x_0<x_1<\cdots<x_n=b$.

$D.$ $\qquad \Delta D_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j].$

$\therefore \quad m_{ij} = \inf\{f(x,y) \mid \forall (x,y) \in \Delta D_{ij}\}.$

$M_{ij} = \sup\{f(x,y) \mid \forall (x,y) \in \Delta D_{ij}\}.$

$\forall \eta_j \in [y_{j-1}, y_j].$ $\qquad \forall x \in [x_{i-1}, x_i].$

$x_i - x_{i-1} = \Delta x_i$

$$\underbrace{\sum_{j=1}^{m}\sum_{i=1}^{n} m_{ij}\Delta x_i \Delta y_j}_{\downarrow\ \iint_D f(x,y)\,dxdy} \le \sum_{j=1}^{m}\left(\sum_{i=1}^{n}\int_{x_{i-1}}^{x_i} f(x,\eta_j)\,dx\right)\Delta y_j \le \underbrace{\sum_{i,j} M_{ij}\Delta x_i \Delta y_j}_{\downarrow\ \iint_D f(x,y)\,dxdy}$$

$$= \sum_{j=1}^{m}\left(\int_a^b f(x,\eta_j)\,dx\right)\Delta y_j = \underbrace{\sum_{j=1}^{m} I(\eta_j)\Delta y_j}_{\downarrow\ \int_c^d I(y)\,dy}$$

$\|T\| \to 0.$

$\max\{\Delta y_j \mid j=1,\cdots,m\} \to 0$

$\min\{\Delta x_i \mid i=1,2,\cdots,n\} \to 0$

$\Longrightarrow \|T\| \to 0$

定理6. 设 $f(x,y) \in R(D)$. $\quad D = [a,b] \times [c,d].$

且设 $\forall x \in [a,b]$. $\quad I(x) = \int_c^d f(x,y)\,dy$ 存在.

则先对 y 后对 x 累次积分 $\int_a^b \left(\int_c^d f(x,y)\,dy\right)dx$ 存在, 且

$$\iint_D f(x,y)\,dxdy = \int_a^b \left(\int_c^d f(x,y)\,dy\right)dx$$

$$= \int_a^b \int_c^d f(x,y)\,dy\,dx$$

$$= \int_a^b dx \int_c^d f(x,y)\,dy$$

$f(x,y) \in C(D).$ $\qquad D = [a,b] \times [c,d].$

则 $\iint_D f(x,y)\,dxdy = \int_a^b\left(\int_c^d f(x,y)\,dy\right)dx = \int_c^d\left(\int_a^b f(x,y)\,dx\right)dy$

定理6. 设 $D = \{(x,y) \mid a \le x \le b,\ y_1(x) \le y \le y_2(x)\}.$ $\quad y_1(x), y_2(x) \in C[a,b].$

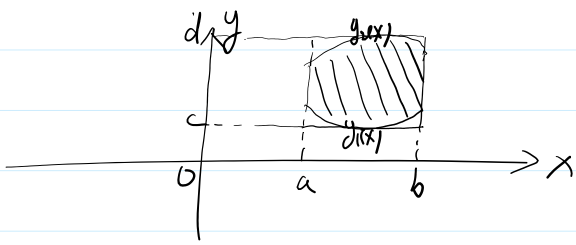
定理6. 设 $D=\{(x,y) \mid a\le x\le b,\ y_1(x)\le y\le y_2(x)\}$. $y_1(x), y_2(x)\in C[a,b]$.
(X-型区域)

且 $f(x,y)\in C(D)$. 则

$$\iint_D f(x,y)\,dxdy = \int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x,y)\,dy\right)dx$$

注: ① 累次积分上限大于下限、
② 外层积分限常数区间.

证:



$c=\min\{y_1(x) \mid x\in[a,b]\}$.

$d=\max\{y_2(x) \mid x\in[a,b]\}$.

$E=[a,b]\times[c,d]\supset D$.

$$g(x,y)=\begin{cases} f(x,y) & (x,y)\in D \\ 0 & (x,y)\in E\setminus D \end{cases}$$

$$\iint_E g(x,y)\,dxdy = \iint_D f(x,y)\,dxdy$$

$$\parallel$$

$$\int_a^b\left(\int_c^d g(x,y)\,dy\right)dx = \int_a^b\left(\int_{y_1(x)}^{y_2(x)} f(x,y)\,dy\right)dx$$

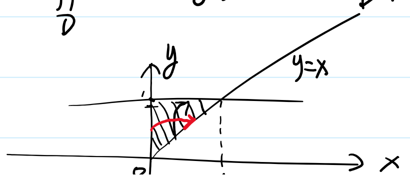
定理7. $D=\{(x,y) \mid c\le y\le d,\ x_1(y)\le x\le x_2(y)\}$, $x_1(y), x_2(y)\in C[c,d]$.
(y-型区域)

设 $f(x,y)\in C(D)$. 则

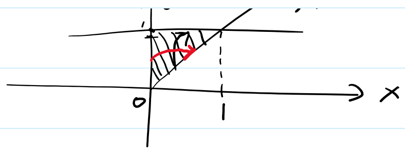
$$\iint_D f(x,y)\,dxdy = \int_c^d\left(\int_{x_1(y)}^{x_2(y)} f(x,y)\,dx\right)dy$$

例1. 计算 $I=\iint_D e^{-y^2}\,dxdy$. D: $x=0$, $y=1$, $y=x$ 围成区域.

解:



解：



$D=\{(x,y)\mid 0\le x\le 1,\quad x\le y\le 1\}$

$I=\int_0^1\left(\int_x^1 e^{-y^2}dy\right)dx$ ✗

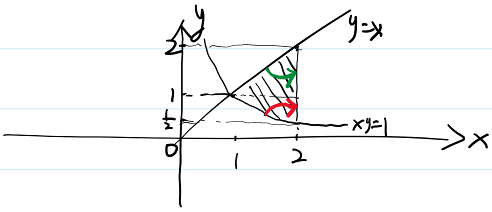
$D=\{(x,y)\mid 0\le y\le 1,\quad 0\le x\le y\}$.

$I=\int_0^1\left(\int_0^y e^{-y^2}dx\right)dy=\int_0^1 e^{-y^2}\cdot y\,dy$

$=-\frac{1}{2}e^{-y^2}\Big|_0^1=\frac{1}{2}(1-e^{-1})$.

例2. 求 $I=\iint\limits_D \frac{x^2}{y^2}dxdy$. D: $x=2$, $y=x$. $xy=1$ 围成.

解:



1°. $D_1=\{(x,y)\mid \frac{1}{2}\le y\le 1,\quad \frac{1}{y}\le x\le 2\}$

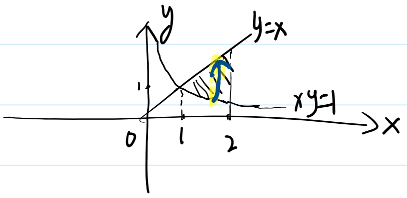
$D_2=\{(x,y)\mid 1\le y\le 2.\quad y\le x\le 2\}$.

$D=D_1\cup D_2$.

$I=\iint\limits_{D_1}\frac{x^2}{y^2}dxdy+\iint\limits_{D_2}\frac{x^2}{y^2}dxdy$

$=\int_{\frac{1}{2}}^1\left(\int_{\frac{1}{y}}^2\frac{x^2}{y^2}dx\right)dy+\int_1^2\left(\int_y^2\frac{x^2}{y^2}dx\right)dy$

2°.



$D=\{(x,y)\mid 1\le x\le 2,\quad \frac{1}{x}\le y\le x\}$.

$I=\int_1^2\left(\int_{\frac{1}{x}}^x\frac{x^2}{y^2}dy\right)dx$

$$= \int_1^2 x^2 \left(-\frac{1}{y}\Big|_{\frac{1}{x}}^{x}\right) dx = \int_1^2 x^2 \left(x - \frac{1}{x}\right) dx$$

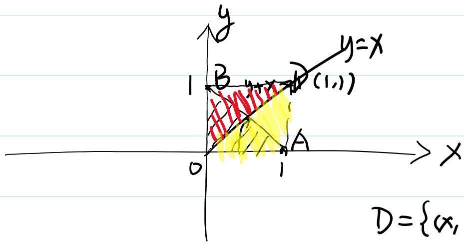
$$= \frac{1}{4}x^4\Big|_1^2 - \frac{1}{2}x^2\Big|_1^2 = \frac{9}{4}.$$

例3. 设 $f(x) \in C[0,1]$. 证明: $\iint\limits_D f(1-y)f(x)\,dxdy = \frac{1}{2}\left(\int_0^1 f(x)\,dx\right)^2$.

D: $(0,0)$, $(1,0)$, $(0,1)$ 为顶点三角形区域.

$$\frac{1}{2}\int_0^1 f(x)\,dx \cdot \int_0^1 f(y)\,dy$$

证:



$D = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le 1-x\}$

$$\iint\limits_D f(1-y)f(x)\,dxdy = \int_0^1 \left(\int_0^{1-x} f(x)f(1-y)\,dy\right) dx$$

$$= \int_0^1 f(x)\left(\int_0^{1-x} f(1-y)\,dy\right) dx$$

$$\overset{1-y=t}{=\!=} \int_0^1 f(x)\left(\int_x^1 f(t)\,dt\right) dx$$

$$= \int_0^1 f(x)\left(\int_x^1 f(y)\,dy\right) dx$$

$$= \iint\limits_{\triangle OBD} f(x)f(y)\,dxdy = \iint\limits_{\triangle OAD} f(x)f(y)\,dxdy$$

$$= \frac{1}{2}\iint\limits_{[0,1]\times[0,1]} f(x)f(y)\,dxdy$$

$$= \frac{1}{2}\int_0^1 f(x)\left(\int_0^1 f(y)\,dy\right) dx$$

$$= \frac{1}{2}\int_0^1 f(x)\,dx \cdot \int_0^1 f(y)\,dy$$

$$= \frac{1}{2}\left(\int_0^1 f(x)\,dx\right)^2.$$

例4. 计算 $I = \iint\limits_D x\left(1 + y f(x^2+y^2)\right) dxdy$.

D: $y = x^3$, $y = 1$, $x = -1$ 围成区域. 且 $f(z) \in C$.

解: